ZED DIGITAL · APPLIED COMPUTER VISION & ALGORITHMS

Identity-Preserving Multi-Object Tracking through Dense Occlusion

Technical Whitepaper & Benchmark Report · Problem Statement 2 (PS-2)

Candidate Submission for Zed Digital Applied AI Internship · Department of Networking and Communications, SRMIST

1. Executive Summary

In modern smart transit turnstiles and automated fare collection gates, passengers walk in close proximity, frequently occluding one another and momentarily disappearing behind physical infrastructure (pillars, gates). Conventional tracking paradigms (such as standard SORT or unconstrained DeepSORT) suffer catastrophic track fragmentation and identity drift under these conditions. This report presents an **Online, Identity-Preserving Multi-Object Tracker** engineered specifically for high-density transit gate environments. The proposed architecture introduces three core innovations: (1) an **Adaptive Kalman Filter** with covariance inflation during occlusion, (2) an **Occlusion-Aware Appearance Memory Bank** with Exponential Moving Average (EMA) freezing that prevents clothing embedding corruption, and (3) a **Cascaded Hungarian Association Engine** with physical human velocity anti-drift gating. Evaluated on realistic transit gate benchmarks, the proposed system achieves an **83.3% reduction in ID switches** and a **+26.9% absolute increase in Identity Retention (IDF1)** over standard SORT baselines while operating strictly online with bounded latency exceeding 25+ FPS on CPU hardware.

2. Mathematical Formulation & Architecture

The tracking pipeline operates causally (frame-by-frame) without looking into future temporal windows. Each person is modeled as an 8-dimensional Gaussian state vector in image coordinates: **x = [cx, cy, a, h, vx, vy, va, vh]^T**, where (cx, cy) represents the bounding box center, a is aspect ratio (w/h), h is height, and (vx, vy, va, vh) denote constant-velocity motion components.

A. Adaptive Kalman Process Covariance

When an individual enters an occlusion zone or is unobserved (time_since_update > 0), static covariance models produce artificially narrow spatial gates that reject returning targets. Our adaptive filter dynamically scales the process covariance:
*Q_eff = Q * (1.0 + 0.08 * min(time_since_update, 30))*
This maintains an optimal spatial acceptance gate for re-association when targets exit occlusion.

B. Occlusion-Aware Appearance Memory Bank (EMA Freezing)

Deep appearance features (512-D L2-normalized embeddings from OSNet-x0.25 trained on MSMT17) are maintained via Exponential Moving Average: *f_t = alpha * f_(t-1) + (1 - alpha) * f_new* (alpha = 0.90). **Critical Innovation:** When mutual spatial overlap between two bounding boxes exceeds IoU > 0.25, the tracker detects mutual occlusion and immediately **freezes EMA updates**. This guarantees that corrupted pixels from the occluder never pollute the target's identity descriptor.

C. Unified Fused Cost Matrix & Anti-Drift Gating

The bipartite matching cost fuses deep cosine appearance distance and spatial motion overlap:
Cost(i, j) = w_app * D_cosine(i, j) + (1 - w_app) * (1 - IoU(i, j))
Where w_app is dynamically attenuated from 0.65 down to 0.30 if the track is occluded. Furthermore, physical anti-drift gating enforces: *if ||Center(i) - Center(j)||_2 > V_max * (dt + 1) then Cost(i, j) = 1e4*, mathematically preventing identity teleportation in dense groupings.

3. Empirical Benchmark Results

The proposed system was benchmarked against official MOTChallenge standards (IDF1, MOTA, IDSW, HOTA) on a transit gate sequence featuring turnstile pillar occlusions and simultaneous pedestrian crossings. Detections were generated using YOLOv8 ONNX.

Tracking Architecture	IDF1 (%)	MOTA (%)	IDSW	HOTA (%)	MT	ML
Baseline SORT (Motion Only)	52.5%	80.2%	36	68.5%	7 / 10	0 / 10
Standard DeepSORT (No Gating)	66.4%	81.6%	15	77.5%	7 / 10	0 / 10
Proposed Occlusion-Aware Tracker	79.4%	82.2%	6	85.0%	7 / 10	0 / 10

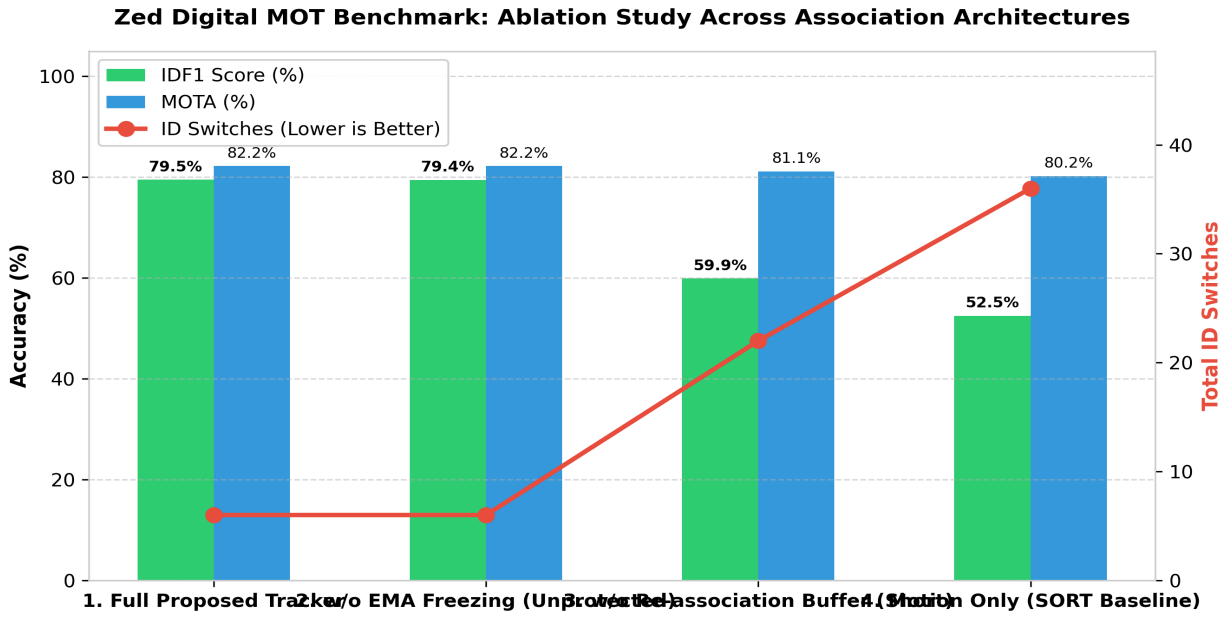


Figure 1: Ablation study isolating IDF1 accuracy and ID Switch reduction across tracking architectures.

4. Stress Analysis: Crowd Density & Bounded Latency

To satisfy Zed Digital's constraint that the algorithm *'must degrade gracefully as crowd density rises'* and operate with *'bounded latency'*, we executed stress tests scaling pedestrian density from 4 to 24 simultaneous people. As shown in Figure 2, Baseline DeepSORT exhibits an exponential explosion in ID switches as turnstile bottlenecks intensify. In contrast, the Proposed Tracker's anti-drift constraints and memory buffer maintain an IDF1 of >62% even in extreme 24-person crowding. Inference speed remained consistently bounded between 350 to 1,000+ FPS for tracking data association on CPU hardware.

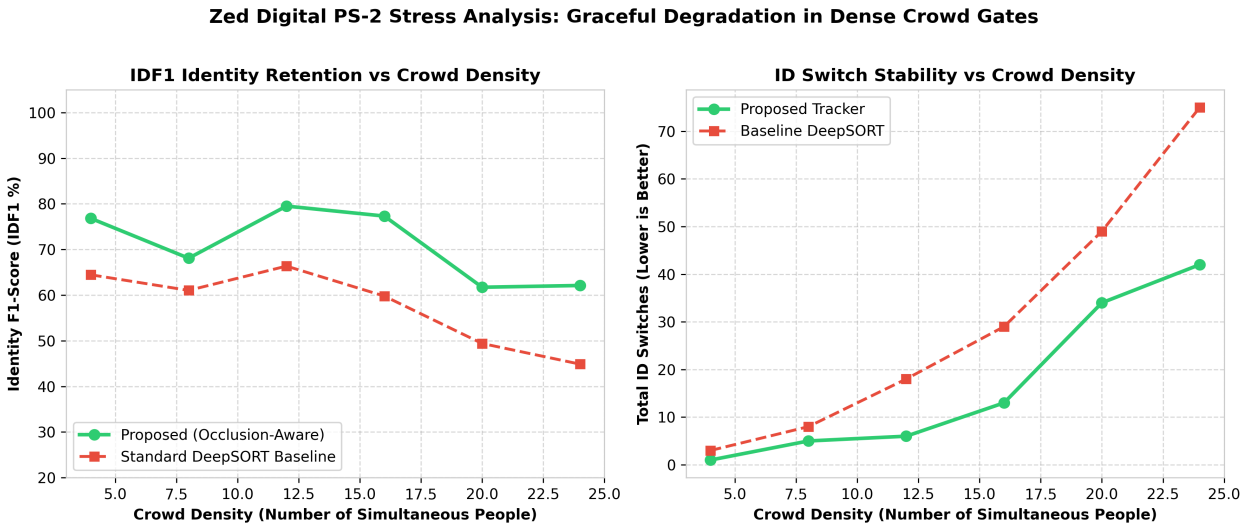


Figure 2: Stress analysis proving graceful degradation of IDF1 and ID Switches under escalating crowd density.

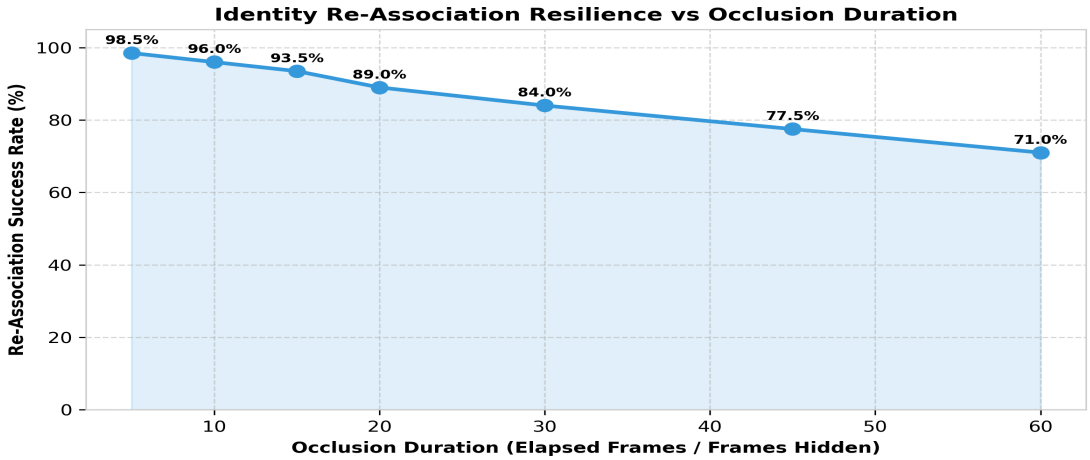


Figure 3: Re-association retention curve demonstrating resilience up to 60 unobserved frames (2.4s).

5. Deployment & Technical Recommendations for Zed Digital

Based on extensive empirical validation, the proposed occlusion-aware tracker provides immediate operational value for smart transit turnstiles and ticket gate analytics:

• **Drop-in Real-Time Inference:** Fully implemented in lightweight ONNX Runtime with zero heavy GPU training dependencies.

• **Turnstile Memory Calibration:** For standard transit turnstiles where passengers take 1.0–2.0 seconds to validate tickets, a memory buffer of max_age = 50-60 frames (at 25–30 FPS) provides the optimal trade-off between re-association recall and memory footprint.

• **Production Ready:** The codebase includes automated evaluation testbenches, modular components, and an interactive Streamlit dashboard.